Roll No. | | | | | | | |

EVEN END SEMESTER EXAMINATION - 2025

Name of the Course: **B.Tech** Semester: **IV**

Name of the Paper: **Solid Mechanics** Paper Code: **TAS 406**

Time: 3 Hours **Maximum Marks: 100**

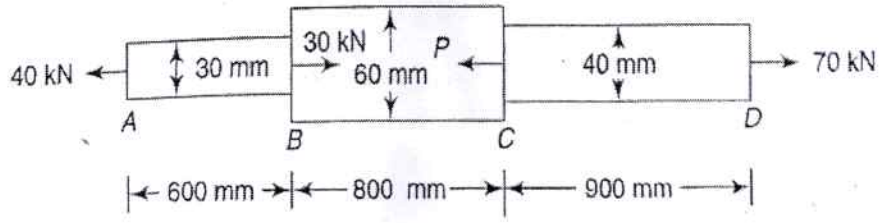
Note:

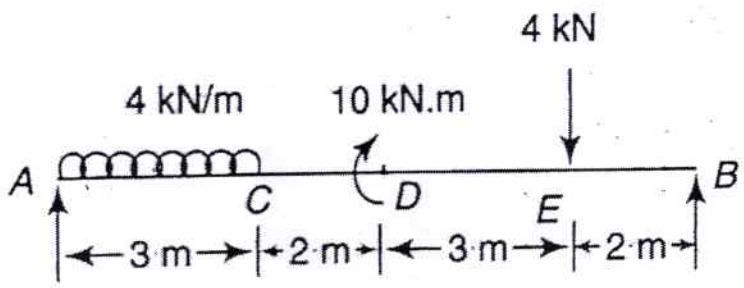
(i) All questions are compulsory

(ii) Answer any two sub questions among a, b &c in each main question

(iii) Total marks in each question are twenty

(iv) Each question carries 10 marks

Q1	**(20 marks)**	
(a)	Explain the following terms: (i) Stress (ii) Young's modulus (iii) Toughness (iv) Poisson's ratio (v) Bulk modulus	
(b)	The stresses on two perpendicular planes through a point in a body are 160 MPa and 100 MPa, both compressive along with a shear stress of 80 MPa. Determine: (i) The normal and shear stresses on a plane inclined at 30° to the plane of 160 MPa stress. Also find the resultant stress and its direction? (ii) The normal stress on a plane at 90° to the inclined plane mentioned in (i).	**CO1, CO2**
(c)	A circular steel bar having three segments is subjected to various forces at different cross-sections as shown in figure. Determine the necessary force to be applied at Section C for the equilibrium of the bar. Also, find the total elongation of the bar. Take E = 202 GPa.  	
Q2	**(20 marks)**	
(a)	A cantilever beam of 10m span carries loads of 4 kN and 6 kN at 2m and 6m respectively from the fixed end along with another load of 6 kN at free end. Draw the shear force and bending moment diagram.	
(b)	A 10-m long simply supported beam carries two-point loads of 10 kN and 6 kN at 2 m and 9 m respectively from the left end. It also has a uniformly distributed load of 4 kN/m run for the length between 4 m and 7 m from the left end. Draw shear force and bending moment diagrams.	**CO3**
(c)	Draw shear force and bending moment diagram for a given beam. Also find the maximum value of bending moment.	



Q3	**(20 marks)**	
(a)	Deduce the bending equation stating the assumptions made.	
(b)	A shaft transmits 280 kW of power at 160 rpm. Determine (i) The diameter of a solid shaft to transmit the required power (ii) The inner and outer diameter of a hollow shaft if the ratio of inner to outer diameter is 2/3 (iii) The percentage saving in the material on using hollow shaft instead of a solid shaft Take the allowable stress as 80 MPa.	**CO4**
(c)	A hollow circular bar as a beam has outside diameter twice of the inside diameter. If it is subjected to a maximum bending moment of 40 kNm and the allowable bending stress is 100 MPa, determine the inside diameter of the bar.	
Q4	**(20 marks)**	
(a)	Derive the expression for slope and deflection at the free end of a cantilever beam subjected to uniformly distributed load over the whole span of the beam.	
(b)	What assumptions are taken in the analysis of thin cylinders? Deduce the expression for circumferential and longitudinal stresses developed in them.	**CO4, CO5**
(c)	A simply supported beam of 8-m length carries two-point loads of 64 kN and 48 kN at 1 m and 4 m respectively from the left-hand end. Find the deflection under each load and the maximum deflection. E =210 GPa and I= 180 x 10^6 mm^4.	
Q5	**(20 marks)**	
(a)	What is meant by crippling or buckling load? Derive the formula for Euler's crippling load for a column subjected to both ends hinged, ends conditions.	
(b)	A thin cylindrical shell 1.2 m long, 500 mm in diameter and 6 mm thick is subjected to an internal pressure of 2 MPa. Calculate: (i) Hoop stress (ii) Longitudinal stress (iii) Change in diameter and length Take E = 200 GPa and Poisson's ratio = 0.3	**CO5, CO6**
(c)	A 4 m long hollow alloy tube with inside and outside diameter as 36 mm and 48 mm elongates by 3 mm under a tensile force of 50 kN. Determine the buckling load for the tube when it is used as a column with both ends pinned and factor of safety of 5.	